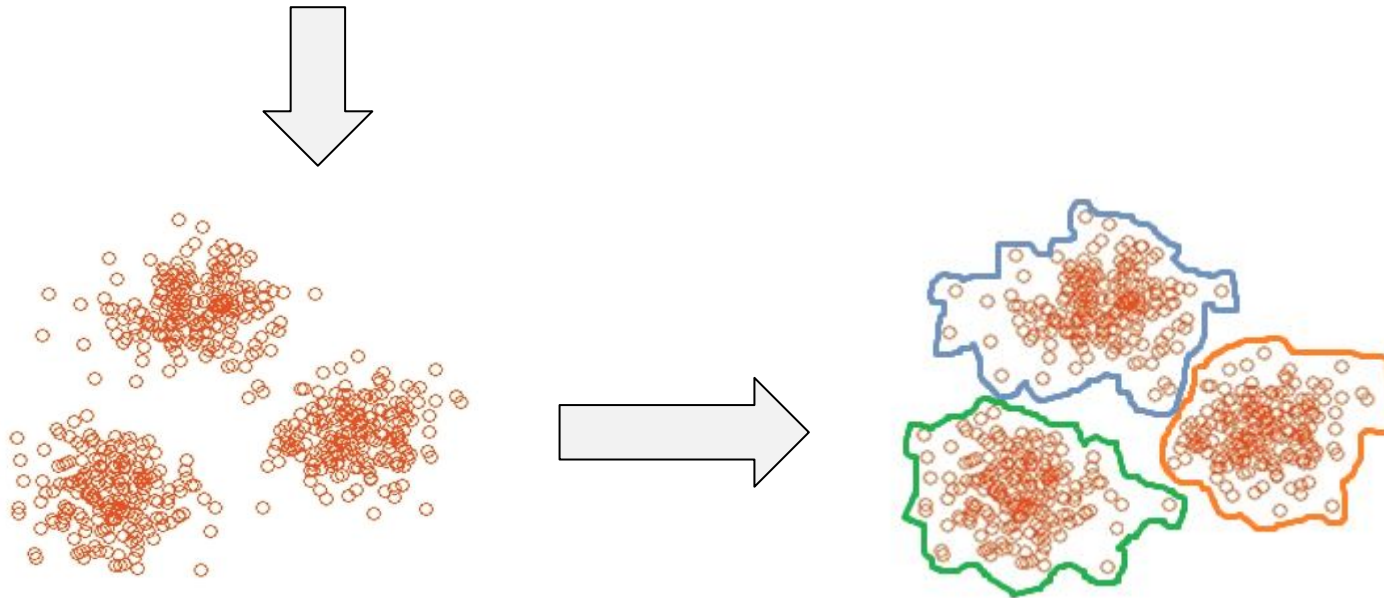


Data Clustering

Problem Definition

ProductID	Name	ProductNumber	Makeflag	FinishedGoodsF...	Color	SafetyStockLevel	ReorderPoint	StandardCost	ListPrice	Size	StockUnitMeasur...	WeightU
1	Adjustable Race	AR-5361	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
2	Bearing Ball	BA-6327	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
3	BB Ball Bearing	BE-2349	True	False	NEGL	800	600	0.0000	0.0000	NEGL	NEGL	NEGL
4	Headset Ball Bea...	BE-2008	False	False	NEGL	800	600	0.0000	0.0000	NEGL	NEGL	NEGL
116	Block	BL-2036	True	False	NEGL	800	600	0.0000	0.0000	NEGL	NEGL	NEGL
117	LL Crankarm	CA-5965	False	False	Black	500	375	0.0000	0.0000	NEGL	NEGL	NEGL
118	ML Crankarm	CA-6736	False	False	Black	500	375	0.0000	0.0000	NEGL	NEGL	NEGL
119	HL Crankarm	CA-7457	False	False	Black	500	375	0.0000	0.0000	NEGL	NEGL	NEGL
120	Chaining Bolts	CB-2903	False	False	Silver	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
121	Chaining Nut	CN-6137	False	False	Silver	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
122	Chaining	CH-7833	False	False	Black	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
123	Crown Race	CR-9961	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
124	Chain Stays	CS-2812	True	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
125	Decal 1	DC-8732	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
126	Decal 2	DC-9624	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
127	Down Tube	DT-2377	True	False	NEGL	800	600	0.0000	0.0000	NEGL	NEGL	NEGL
128	Mountain End Caps	EC-9092	True	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
129	Road End Caps	EC-9096	True	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
130	Touring End Caps	EC-7209	True	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
131	Fork End	FE-3760	True	False	NEGL	800	600	0.0000	0.0000	NEGL	NEGL	NEGL
132	Freehub	FH-2981	False	False	Silver	500	375	0.0000	0.0000	NEGL	NEGL	NEGL
141	Flat Washer 1	FW-1000	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
142	Flat Washer 6	FW-1200	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
143	Flat Washer 2	FW-1400	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
144	Flat Washer 9	FW-3400	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
145	Flat Washer 4	FW-3800	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
146	Flat Washer 3	FW-5160	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
147	Flat Washer 8	FW-5800	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
148	Flat Washer 5	FW-7160	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
149	Flat Washer 7	FW-9160	False	False	NEGL	1000	750	0.0000	0.0000	NEGL	NEGL	NEGL
150	Fork Crown	FC-3854	True	False	NEGL	800	600	0.0000	0.0000	NEGL	NEGL	NEGL

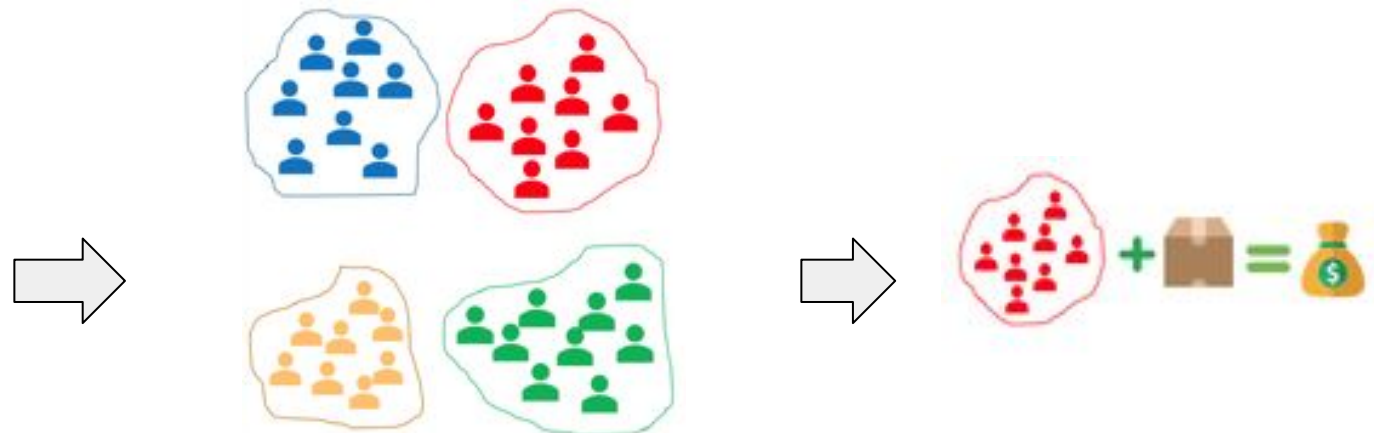
- Identify a finite set of categories or clusters to describe the data so that similar data points belong to the same cluster.



where inter-clusters are similar
while intra-clusters are dissimilar

Problem Definition

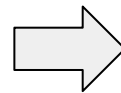
	Meat	Milk	Grocery	Frozen	Detergents
0	12669	9656	7561	214	2674
1	7057	9810	9568	1762	3293
2	6353	8808	7684	2405	3516
3	13265	1196	4221	6404	507
4	22615	5410	7198	3915	1777
5	9413	8259	5126	666	1795
6	12126	3199	6975	480	3140
7	7579	4956	9426	1669	3321
8	5963	3648	6192	425	1716
9	6006	11093	18881	1159	7425
10	3366	5403	12974	4400	5977
11	13146	1124	4523	1420	549
12	31714	12319	11757	287	3881
13	21217	6208	14982	3095	6707
14	24653	9465	12091	294	5058
15	10253	1114	3821	397	964
16	1020	8816	12121	134	4508
17	5876	6157	2933	839	370
18	18601	6327	10099	2205	2767
19	7780	2495	9464	669	2518



Identifying the potential customer base for selling the product

Applications - Documents Clustering

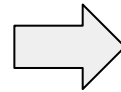
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2	6353	8808	7684	2405	3516
3	13265	1196	4221	6404	507
4	22615	5410	7198	3915	
5	9413	8259	5126	666	
6	12126	3199	6975	480	
7	7579	4956	9426	1669	
8	5963	3648	6192	425	
9	6006	11093	18881	1159	
10	3366	5403	12974	4400	
11	13146	1124	4523	1420	
12	31714	12319	11757	287	
13	21217	6208	14982	3095	6007
14	24653	9465	12091	294	5058
15	10253	1114	3821	397	964
16	1020	8816	12121	134	4508
17	5876	6157	2933	839	370
18	18601	6327	10099	2205	2767
19	7780	2495	9464	669	2518



Cluster documents in multiple categories

Applications - City Planning

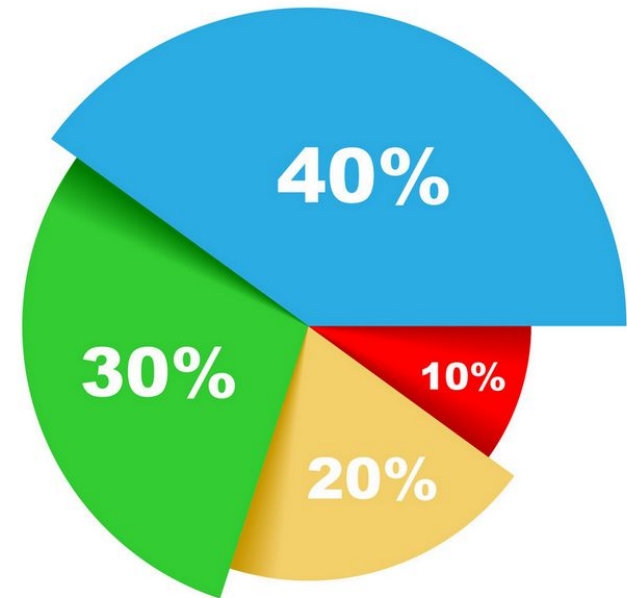
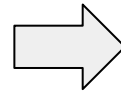
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2	6353	8808	7684	2405	3516
3	13265	1196	4221	6404	507
4	22615	5410	7198	3915	1777
5	9413	8259	5126	666	1795
6	12126	3199	6975	480	3140
7	7579	4956	9426	1669	3321
8	5963	3648	6192	425	1716
9	6006	11093	18881	1159	7425
10	3366	5403	12974	4400	5977
11	13146	1124	4523	1420	549
12	31714	12319	11757	287	3881
13	21217	6208	14982	3095	6707
14	24653	9465	12091	294	5058
15	10253	1114	3821	397	964
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18	18601	6327	10099	2205	2767
19	7780	2495	9464	669	2518



Groups of houses according to their house type, value and geographical location.

Applications - WebLogs

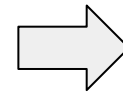
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4	22615	5410	7198	3915	
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6	12126	3199	6975	480	
7	7579	4956	9426	1669	
8	5963	3648	6192	425	
9	6006	11093	18881	1159	
10	3366	5403	12974	4400	
11	13146	1124	4523	1420	
12	31714	12319	11757	287	3881
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15	10253	1114	3821	397	964
16	1020	8816	12121	134	4508
17	5876	6157	2933	839	370
18	18601	6327	10099	2205	2767
19	7780	2495	9464	669	2518



Clustering weblog data to discover groups of similar access patterns

Applications - Library

	F1	F2	F3	F4	F5
0	12669	9656	7561	214	2674
1	7057	9810	9568	1762	3293
2	6353	8808	7684	2405	3516
3	13265	1196	4221	6404	507
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13	21217	6208	14982	3095	6707
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15	10253	1114	3821	397	964
16	1020	8816	12121	134	4508
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19	7780	2495	9464	669	2518

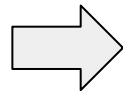


Clustering books into groups

Data Clustering

Balance Purchases CashAdv CrdtLine

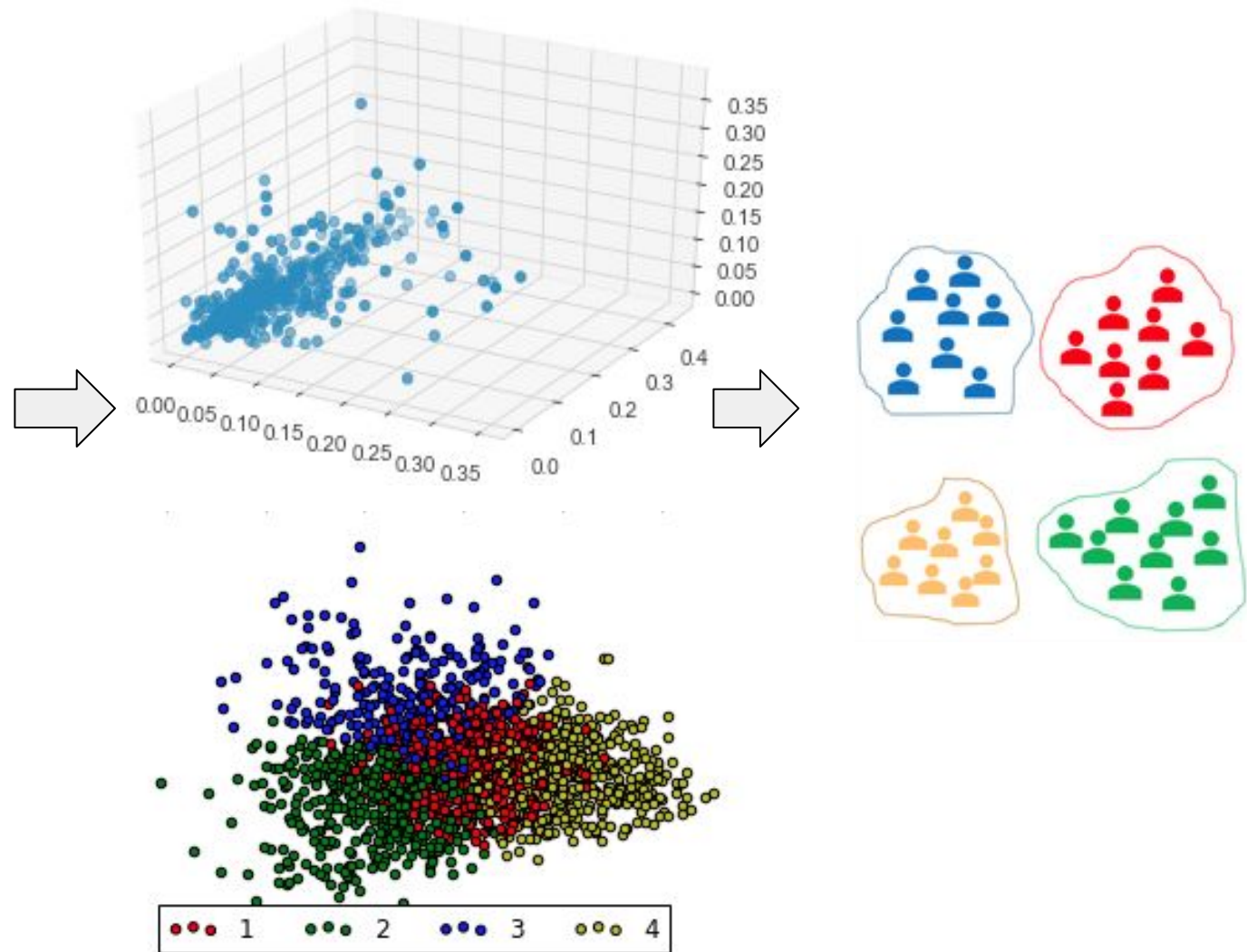
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15	10253	1114	3821	964
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18	18601	6327	10099	2767
19	7780	2495	9464	2518



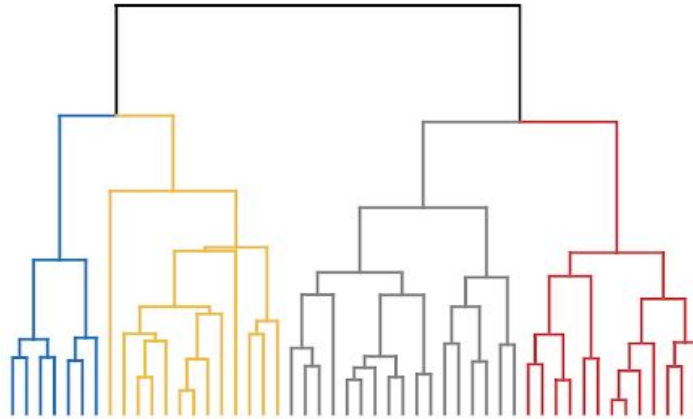
Make meaningful insights from the data

Data Clustering Techniques

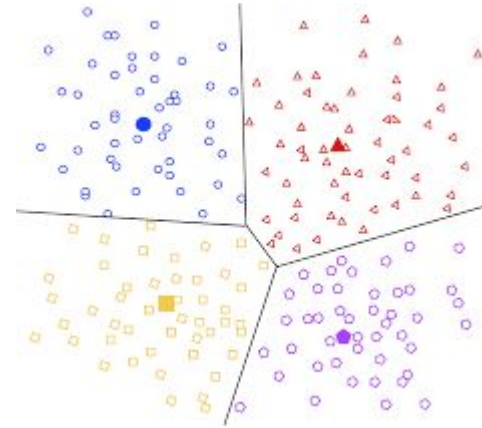
	Meat	Milk	Grocery	Frozen	Detergents
0	12669	9656	7561	214	2674
1	7057	9810	9568	1762	3293
2	6353	8808	7684	2405	3516
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9	6006	11093	18881	1159	7425
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11	13146	1124	4523	1420	549
12	31714	12319	11757	287	3881
13	21217	6208	14982	3095	6707
14	24653	9465	12091	294	5058
15	10253	1114	3821	397	964
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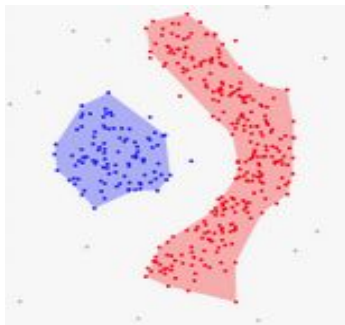
Data Clustering Techniques



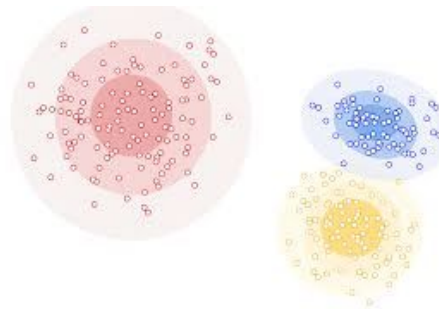
Hierarchical clustering



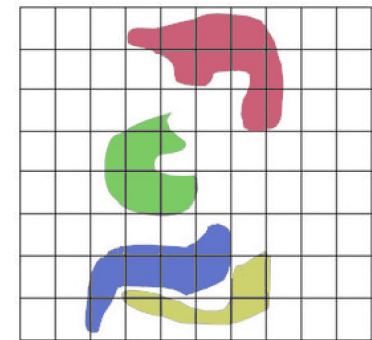
Centroid-based clustering



Density-based clustering

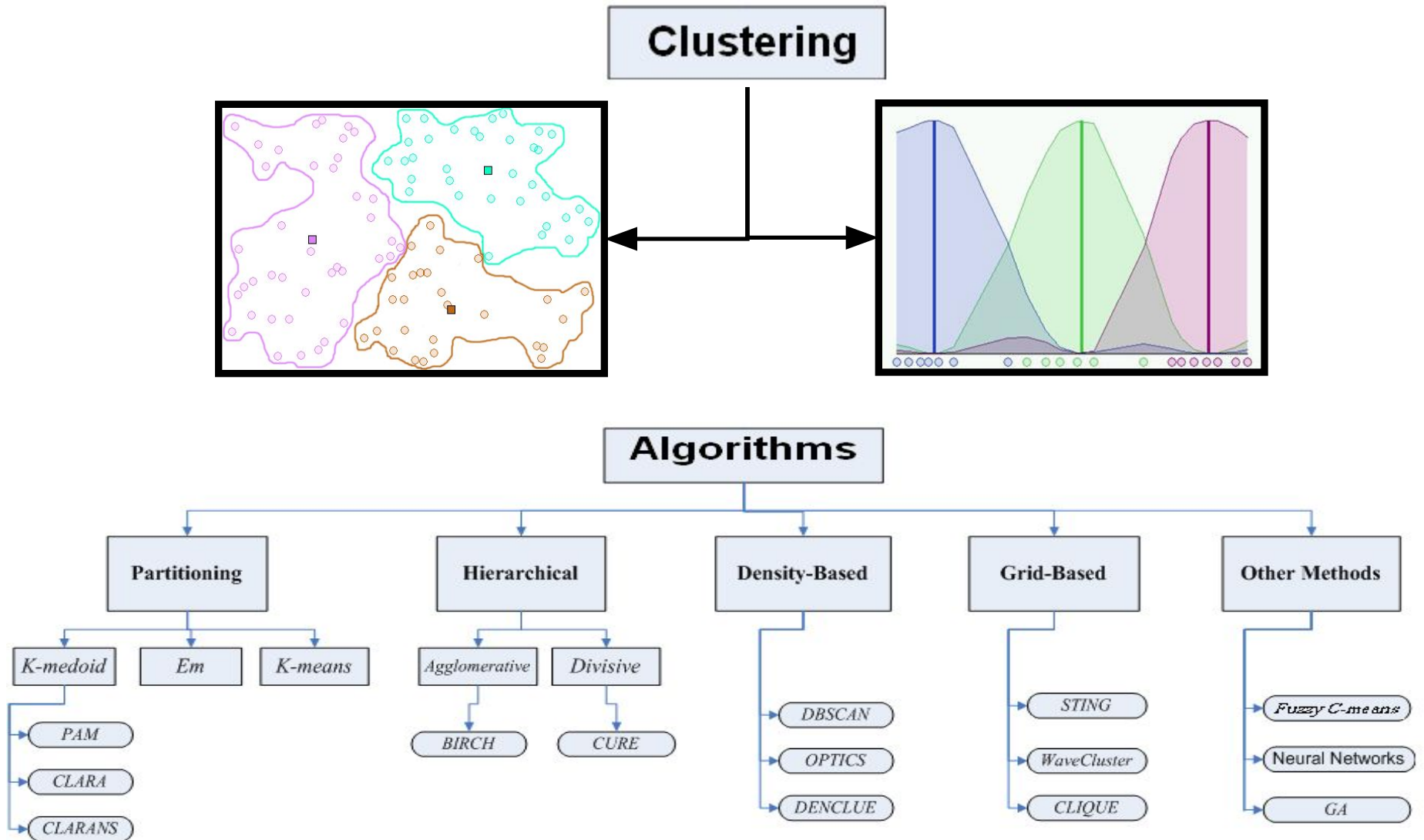


Distribution-based clustering

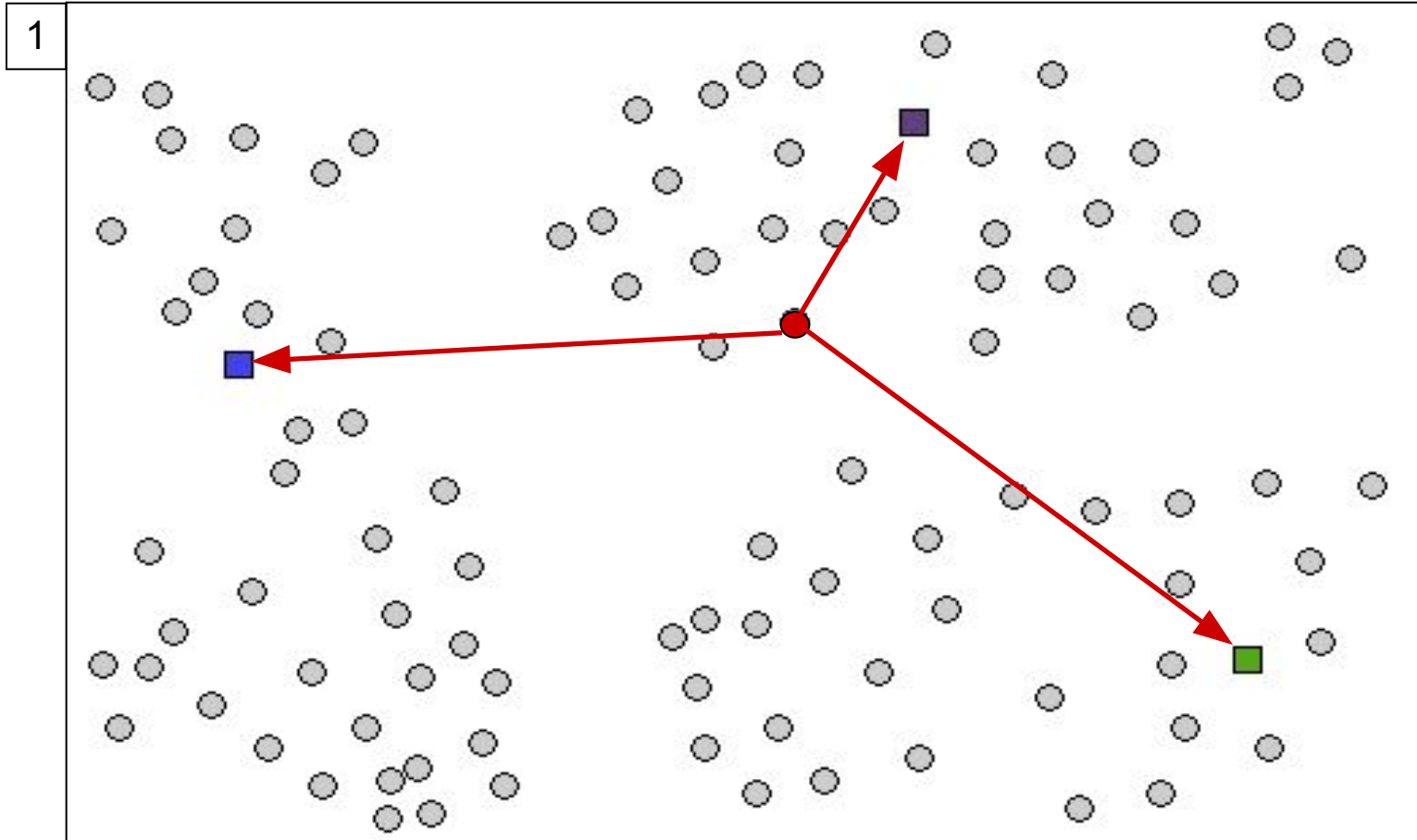


Grid-based clustering

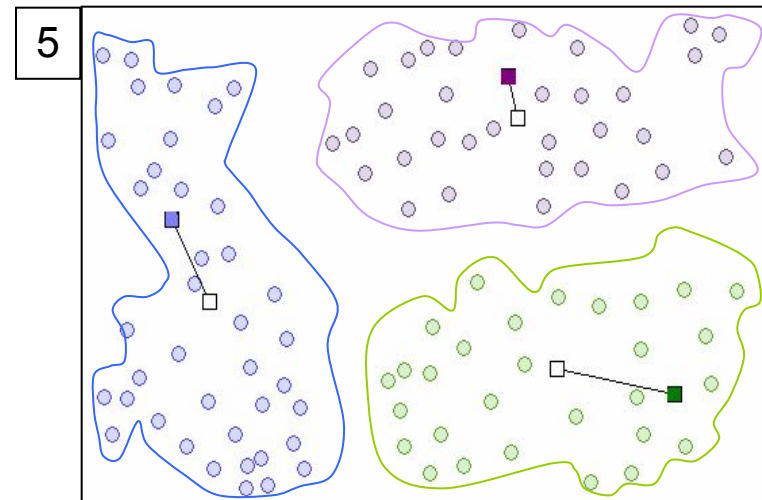
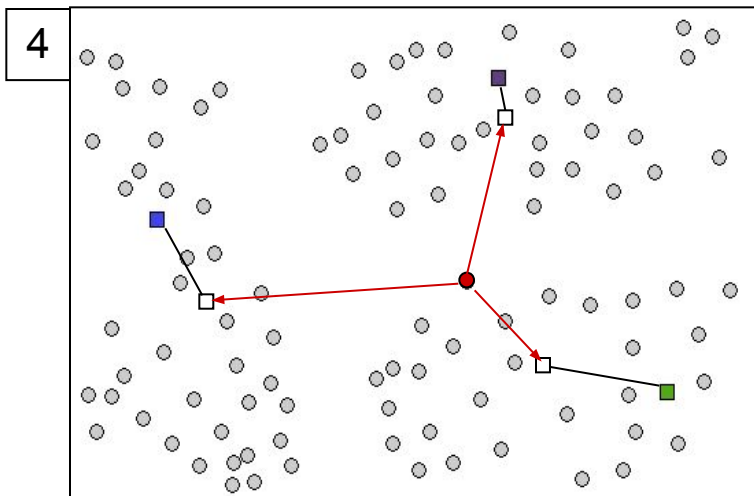
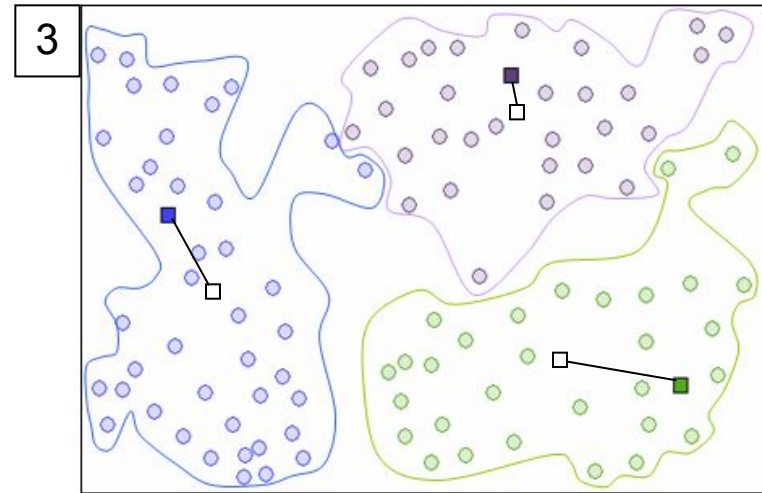
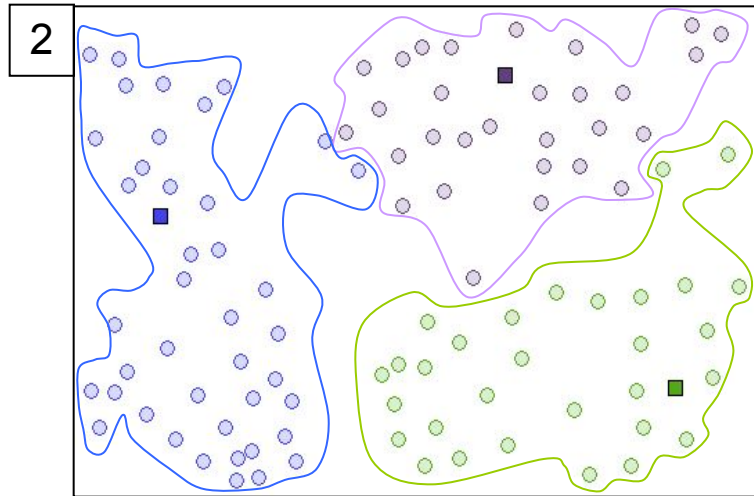
Data Clustering Techniques



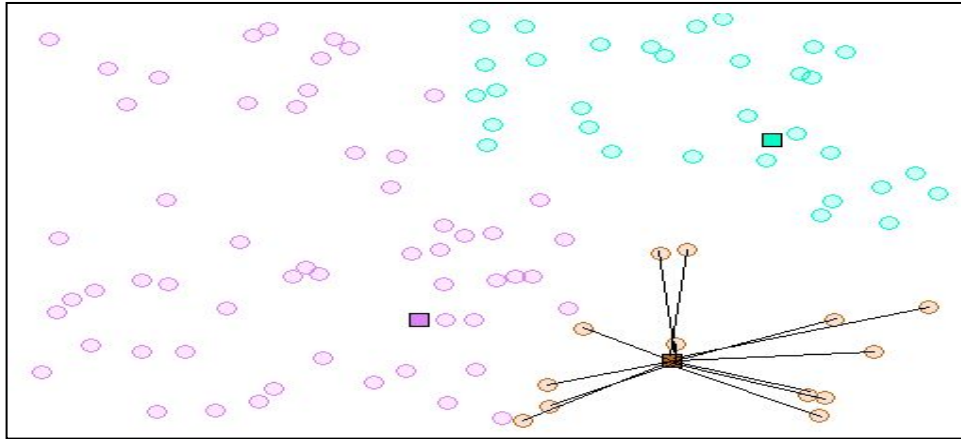
K-means Algorithm



K-means Algorithm



K-means



The main aim is to find groups of high quality where inter-clusters are similar while intra-clusters are dissimilar

$$J = \sum_{j=1}^k \sum_{i=1}^n \underbrace{\|x_i^{(j)} - c_j\|^2}_{\text{Distance function}}$$

number of clusters k

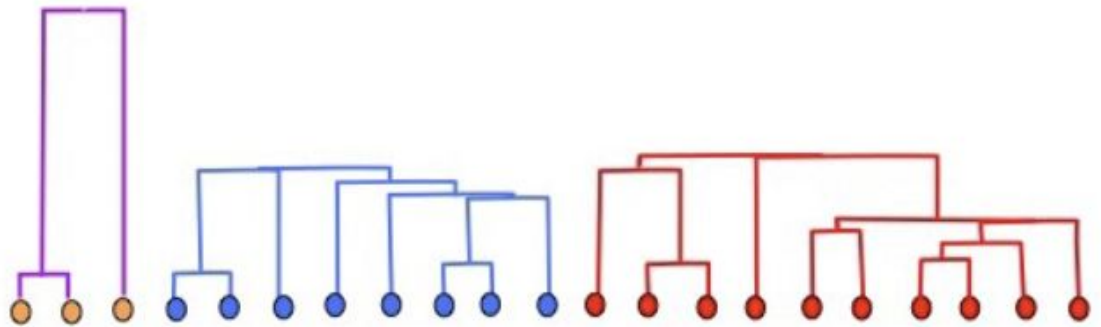
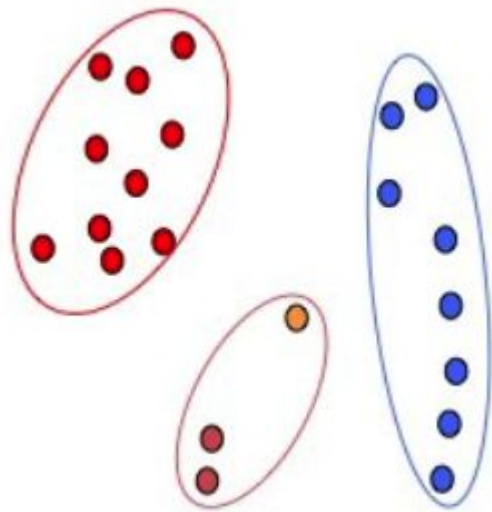
number of cases n

case i

centroid for cluster j

Best Clustering = min (J) = max(1/J)

Single-linkage



GMM

